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RESEARCH ARTICLE

Integrating passive acoustic monitoring and point counts to assess occupancy of secretive marsh birds in Tennessee

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Abstract

Monitoring of wildlife populations is fundamental for evidence-based conservation yet remains particularly challenging for cryptic species. Secretive marsh birds (hereafter marsh birds) are notoriously difficult to monitor because of their shy, unobtrusive nature and dense vegetation preferences. Existing approaches, such as call-broadcast point count surveys, can result in low detection probabilities and increased uncertainty in population assessments. We integrated passive acoustic monitoring with traditional call-broadcast point count surveys to evaluate how differences in survey design influence estimates of occupancy and detection for marsh birds. We further used dynamic occupancy models to examine how local and landscape characteristics influence patterns of site use while accounting for potential temporal changes in occupancy. During April–May 2022–2023, we conducted point count surveys at 78 wetland points in Tennessee, USA, for 6 marsh bird species including the king rail (*Rallus elegans*), least bittern (*Ixobrychus exilis*), common gallinule (*Gallinula galeata*), American coot (*Fulica americana*), pied-billed grebe (*Podilymbus podiceps*), and sora (*Porzana carolina*). Prior to point counts, we deployed autonomous recording units (ARUs) at each point and recorded for 5-day periods during times when marsh birds are most active (i.e., 3 hr after sunrise, 3 hr before sunset). We rarely detected our focal species using point count surveys; however, passive acoustic

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monitoring with ARUs increased detection probabilities, consistent with greater temporal sampling effort. Furthermore, we found that landscape characteristics were stronger predictors of marsh bird site use than local vegetation characteristics in Tennessee. Generalist species such as American coots, pied-billed grebes, and soras occurred statewide, seemingly more tolerant of agricultural cover and forested wetlands. In contrast, king rails were rarely detected (1.3% of our surveys), suggesting current wetland conditions in Tennessee may not support breeding populations of this marsh specialist. Collectively, our study demonstrates that monitoring methodology can strongly influence estimates of occupancy and detection in low-detection systems, and that differences among survey approaches are closely linked to temporal sampling effort and species-specific detection processes. However, challenges in estimating occupancy of rare species at the edges of their ranges remain, and careful cost-benefit considerations are warranted when incorporating passive acoustic monitoring into long-term programs.

KEYWORDS

autonomous recording units, occupancy, passive acoustic monitoring, point counts, secretive marsh birds, Tennessee, wetlands

Secretive marsh birds (hereafter marsh birds), including rails, bitterns, gallinules, and grebes, are challenging to monitor, and estimating population vital rates remains difficult (Lévêque et al. 2021). Interest in marsh birds has grown because of perceived declines and habitat loss, yet basic biological information and population indices are still lacking for some species (Lévêque et al. 2021). Occupancy estimates have become an important tool for monitoring population status and distribution, especially for rare and elusive species like marsh birds (MacKenzie et al. 2002). However, concerns persist as occupancy models remain difficult to implement because of technical obstacles like limited personnel, equipment, and wetland access (Conway 2011). The North American Marsh Bird Monitoring Program (NAMBMP) provides a standardized approach to monitor marsh bird populations and estimate occupancy but may yield low detection probabilities for some species, limiting inference in certain systems (Fournier et al. 2022).

Because occupancy estimates depend on the survey methodology used to collect detection data, interest has grown in complementary methods to improve detection and strengthen inference. For example, use of passive acoustic monitoring for remote wildlife surveys has recently increased to supplement marsh bird monitoring efforts. Autonomous recording units (ARUs) are acoustic monitoring devices designed to record external sounds (Shonfield and Bayne 2017) and can efficiently detect vocalizing avian species (Tegeler et al. 2012, Venier et al. 2012, Holmes et al. 2014) and improve marsh bird detections given they are present (Bobay et al. 2018, Schroeder and McRae 2020, Znidersic et al. 2020). For example, ARUs doubled the detections of the federally threatened eastern black rail (*Laterallus jamaicensis*) in North Carolina, USA, compared to point counts (Bobay et al. 2018). However, detection outcomes can vary among species and survey approaches, as some marsh birds respond weakly to call-broadcast surveys or vary in vocalization

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behavior (Conway and Gibbs 2005, Conway and Nadeau 2010). Bobay et al. (2018) also found that ARU data added fewer detections for least bitterns (*Ixobrychus exilis*). In some cases, ARU-based counts have performed similarly to in-person point count surveys in the Canadian provinces of Saskatchewan, Ontario, Prince Edward Island, New Brunswick, and Nova Scotia (Stewart et al. 2020). These findings suggest that differences regarding survey methodologies may influence detection estimates and occupancy, with implications for monitoring program designs.

Occupancy estimates are derived from detection histories and therefore depend not only on ecological processes but also the survey methodologies used to collect detection data (MacKenzie et al. 2002). Although occupancy models account for imperfect detection, resulting estimates remain influenced by survey design characteristics such as temporal sampling effort, spatial coverage, and species-specific detection processes (Kéry and Schmidt 2008, Kellner and Swihart 2014). Consequently, disparate monitoring methodologies applied to the same system may generate different estimates of occupancy and detection probability, leading to different ecological inferences. These considerations are particularly relevant for secretive marsh birds, where low detection rates, logistical constraints, and variation in vocal behavior can strongly influence observed detection histories (Conway 2011, Fournier et al. 2022). Understanding how commonly used monitoring methodologies influence inference is therefore important for interpreting occupancy estimates and informing monitoring programs.

The population status, geographic distribution, and habitat associations of marsh birds in Tennessee, USA, are largely unknown, though the state may provide important stop-over habitat and occasional breeding habitat (Webb et al. 2022). As of 2024, least bitterns and king rails (*Rallus elegans*) are listed as Deemed in Need of Management in Tennessee (Withers 2016). Historically, least bitterns, king rails, and common gallinules (*Gallinula galeata*) bred in Tennessee (Wetmore 1939). American bittern (*Botaurus lentiginosus*) and sora (*Porzana carolina*) migrate through Tennessee (Wetmore 1939), whereas the American coot (*Fulica americana*) is a common resident, but few nests have been recorded in the state (Wetmore 1939, Fink et al. 2023). Least bitterns were once found nesting in vegetation communities dominated by giant cutgrass (*Zizaniopsis miliacea*) at Reelfoot Lake, Lake County in northwest Tennessee, although increasing prevalence of woody vegetation in the Reelfoot area may have decreased breeding densities (Winstead 2004, Winstead and King 2006).

Previous studies indicate that persistent emergent vegetation (e.g., *Typha* spp.), horizontal cover, and presence of shallow water are important indicators for marsh bird habitat use (Lor and Malecki 2006). Interspersion of marsh area and water is also an important indicator of marsh bird abundance and nesting (Rehm and Baldassarre 2007). Generally, emergent vegetation (Stevens and Conway 2020), water-vegetation interspersion (Rehm and Baldassarre 2007), and wetland complexity (Guadagnin and Maltchik 2006) influence marsh bird habitat selection in other geographies.

Our first objective was to evaluate how different monitoring methodologies commonly used to monitor marsh birds influence estimates of detection and probability of use. Specifically, we compared inference derived from call-broadcast point count (PC) surveys, passive acoustic monitoring using ARUs (ARU), and a combined approach (PC + ARU) fit to single-season occupancy models. Because these survey methods differed substantially in temporal sampling effort and detection processes, we did not aim to identify a superior method but to evaluate how commonly used survey designs influenced ecological inference. We expected that survey approaches with greater temporal sampling effort (e.g., ARUs) would yield higher detection probabilities and reduced uncertainty in occupancy estimates. Our second objective was to investigate the influence of local- and landscape-scale variables on detection probability and occupancy for 5 marsh bird species in Tennessee using a survey design that provided sufficient temporal replication to support dynamic occupancy modeling. By using the combined PC + ARU dataset, we leveraged repeated observations within survey periods to evaluate how habitat characteristics influenced patterns of site use in a temporally dynamic system. Previous studies have found that marsh bird occupancy is associated with landscape cover, so we

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predicted that landscape-scale factors would have a stronger influence on occupancy than local vegetation characteristics (Bradshaw et al. 2020).

STUDY AREA

We conducted fieldwork on public lands throughout western and central Tennessee, USA. In 2022, we surveyed wetlands in western Tennessee within the watersheds of the Obion, Hatchie, Forked Deer, and lower Mississippi River watersheds (Figure 1). In 2023, we surveyed central Tennessee from the Tennessee River to the edge of the Cumberland Plateau (Figure 1). Survey sites included a range of wetland sizes (0.5–20.2 ha; mean = 15.0 ha) and an elevation gradient of approximately 95–250 m above sea level. Spring temperatures in Tennessee ranged from 4°C to 25°C, and mean annual precipitation was approximately 140 cm.

Historically, western Tennessee was characterized by bottomland hardwood forests, but intense river channelization and floodplain modification altered hydrologic processes and reduced natural sediment deposition (Johnson 2007). Today, much of western Tennessee has been converted to agriculture and predominantly consists of corn, wheat, soybeans, and cotton (U.S. Fish and Wildlife Service 2006*a*, *b*; 2014). Central Tennessee was characterized by central hardwood forests, predominantly a mix of oak-hickory (*Quercus* spp. and *Carya* spp.) stands and higher elevations.

The dominant wetland type throughout Tennessee consists of bottomland hardwood forests that are periodically inundated by flood events or heavy rainfall (U.S. Fish and Wildlife Service 2006*a*, *b*). Common overstory species include water oak (*Quercus nigra*), sweetgum (*Liquidambar styraciflua*), and bald cypress (*Taxodium distichum*), whereas mid- and understory vegetation includes buttonbush (*Cephalanthus occidentalis*) and black willow (*Salix nigra*). Emergent herbaceous marshes are uncommon throughout the state but are primarily dominated by giant cutgrass, swamp loosestrife (*Decodon verticillatus*), and common cattail (*Typha* spp.; U.S. Fish and Wildlife Service 2006*a*). As a result of extensive historical land use change and river channelization, many wetlands in Tennessee are relatively small, fragmented, and intensively managed, and may

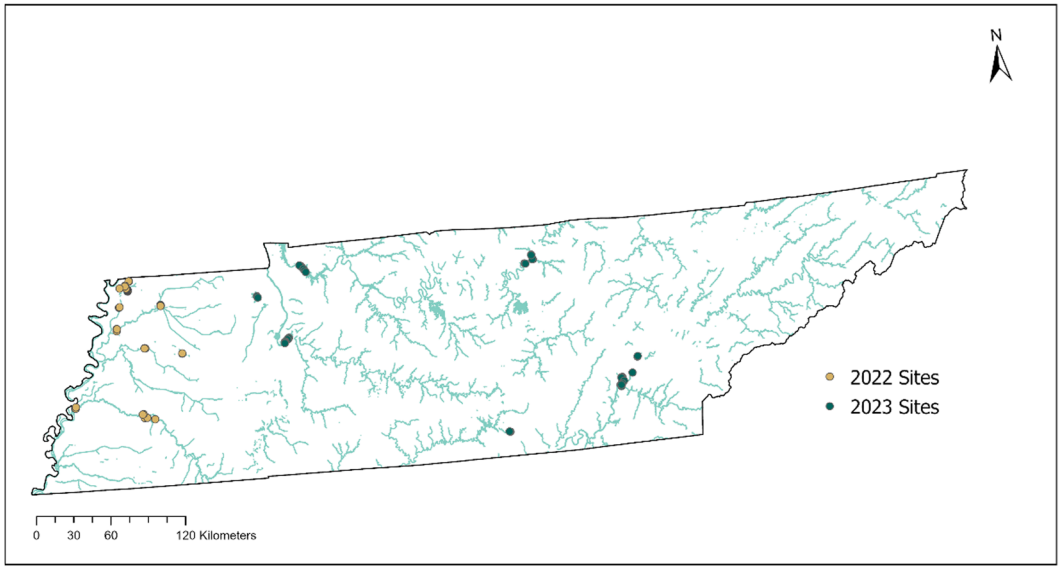


FIGURE 1 Study sites split by year to determine occupancy status of secretive marsh birds in western and central Tennessee, USA, during 15 April–31 May 2022–2023.

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function primarily as migratory stopover habitat rather than supporting consistent breeding populations of marsh birds.

METHODS

Site selection

We limited potential sites to public lands managed by the Tennessee Wildlife Resources Agency and U.S. Fish and Wildlife Service. We identified potential survey sites using a combination of spatial raster data from the 2019 National Land Cover Database (NLCD; Dewitz 2021), the National Wetlands Inventory (NWI; U.S. Fish and Wildlife Service 2004), and the Landfire Vegetation Type database (LANDFIRE 2016). We first selected habitat attributes that potentially influence marsh bird occupancy (e.g., emergent vegetation) to create a single raster layer representing potential emergent marsh habitat and therefore possible survey sites (Table S1).

To achieve broad geographic coverage while maintaining a spatially balanced design, we generated 80 candidate sites in western Tennessee in 2022 and 80 candidate sites in central Tennessee in 2023 using the spsurvey R package (Dumelle 2023). Candidate sites were randomly selected from the emergent marsh raster layer using a generalized random tessellation stratified spatially balanced survey design (GRTS; Johnson et al. 2009). We constrained sites to be ≥1,500 m apart to ensure spatial independence among sites. We ground-truthed all candidate wetland sites in March of 2022 and 2023 and only retained sites containing emergent vegetation and wetland diameter ≥400 m, consistent with marsh bird survey recommendations (Conway 2011). Within each wetland site, we randomly selected 1–3 survey points depending on wetland size. Survey points were required to be ≥400 m apart and were randomly located within suitable emergent marsh habitat (Figure 1; Conway 2011).

Survey methods

PC surveys

We surveyed marsh birds following the NAMBMP (Conway 2011). From 15 April–31 May 2022–2023, we conducted call-broadcast PC surveys at each survey point 3 times over 3 2-week survey periods (Conway 2011). There was a median of 15 days (min. = 14, max. = 16) between PC surveys. Prior to each survey, we had a 10-minute quiet period after arriving at the survey location to limit disturbance. Surveys began with a 5-minute passive listening period followed by a 1-minute audio broadcast of each focal species' calls using electronic game callers (FOXPRO Inferno Caller, All Predator Calls, St. George, UT, USA). Each 1-minute period consisted of 30 seconds of vocalizations followed by 30 seconds of silence during which observers listened to calls and scanned for visual detections of marsh birds. The order of calls began with the least intrusive species and ended with the most intrusive (in order: least bittern, king rail, American coot, common gallinule, sora, pied-billed grebe; Conway 2011). We grouped detections into distance bands in increments of 10 m, with a maximum distance of 100 m, beyond which we did not include detections. Our focal species included least bittern, king rail, common gallinule, American coot, and pied-billed grebe (*Podilymbus podiceps*) based on the recommendations of the NAMBMP (Conway 2011). We opportunistically included migrating sora in our sampling effort because of a high number of detections and uncertainty in their migratory timing or habitat use in Tennessee. At the start of every call-broadcast PC survey, we recorded variables that could potentially affect marsh bird detection including wind speed (km/hr), background noise level (0–5), air temperature (°C), cloud cover, and precipitation (Conway et al. 2004, Nadeau et al. 2008). We did not conduct surveys if sustained precipitation, fog, or high wind speeds (>20 km/hr) were present.

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ARU surveys

We deployed ARUs at every survey point 5 days before the call-broadcast survey for each of our 3 survey visits. We used Song Meter SM4 and Song Meter Mini units, which have comparable recording quality (Wildlife Acoustics Inc., Maynard, MA, USA). We programmed ARUs to record for 5 days and stopped the recording after the call-broadcast PC survey. We programmed ARUs to record during hours when marsh birds are most active: 30 minutes before sunrise to 3 hours after sunrise and 3 hours before sunset to 30 minutes after sunset (Conway 2011). We mounted ARUs to T-posts approximately 1.5 m above the ground.

Manual processing of ARU data is a time-intensive process, and thus automated annotation of bird calls may be a faster and less costly alternative to manual annotation (Cole et al. 2022). We used BirdNET automated classifier software developed by the Cornell Lab of Ornithology (Ithaca, NY, USA; Kahl et al. 2021); BirdNET is a deep neural network that can identify songs from >900 species of North American birds (Kahl et al. 2021). We set the confidence threshold low (0.20), which resulted in more false positives than false negatives. Owing to the large volume of false positives produced by BirdNET (68%), we manually verified all detections by visual scanning of spectrograms in Raven Pro version 1.6 (Cornell Lab of Ornithology; Charif et al. 2010) and auditory confirmation by trained observers. We marked true detections of marsh birds with 1 and false positives (e.g., non-target sounds marked by the detector) with 0. Because ARU and PC surveys differ substantially in temporal sampling effort and detection processes, these approaches generate data with different structures that can influence the probability of detecting occurrence and downstream inference.

Data analyses

Objective 1: Comparison between ARUs and PCs

To evaluate how different survey approaches (i.e., PC, ARU, and combined PC + ARU) influence estimates of occupancy for focal marsh bird species in Tennessee, we used single-season occupancy models (MacKenzie et al. 2002). We divided the sampling season (April–May) into 3 survey periods, each approximately 2 weeks apart. For the PC dataset, we conducted one 11-minute PC survey during each survey period, resulting in 3 total survey occasions per site. For the ARU dataset, each survey period consisted of 5 consecutive days of recordings, with each day treated as a separate survey occasion, resulting in 15 total survey occasions per site (5 days × 3 survey periods). For the combined PC + ARU dataset, detections from the PC surveys were incorporated into the corresponding ARU sampling period by assigning a detection (1) to the day on which the PC survey occurred on the last day of each 5-day ARU sampling window. We assigned a 0 to all other days without detections, resulting in binary encounter histories, where each element represents a survey occasion and 1 indicates detection and 0 indicates non-detection.

We estimated occupancy (ψ) and probability of detecting occupancy (p) for all focal species using the unmarked package (Fiske and Chandler 2011, Kellner et al. 2023). We used single-season occupancy models for all datasets to maintain a consistent modeling framework across survey approaches and to evaluate how inference varies under different sampling designs. Because we had 1–3 points within each site, we evaluated whether a random effect for site was appropriate to account for non-independence of survey points. However, models including site as a random effect had poor convergence, likely due to most sites having one point so we did not include a random effect of site in the final models. We compared model estimates across survey approaches (PC, ARU, and PC + ARU) by examining 85% confidence intervals and quantifying uncertainty in occupancy estimates using the coefficient of variation (CV) and confidence interval width on the probability scale (Reed et al. 2002, Arnold 2010). Survey approaches differed substantially in temporal sampling effort and spatial detection extent. The ARUs recorded approximately 7 hours per day over 5 days per survey period (~105 hours per point) and were not constrained by a

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fixed detection radius, whereas PC surveys totaled approximately 30 minutes per point across 3 visits. As such, differences in probability of detecting occupancy and occupancy estimates among survey approaches are expected to reflect differences in sampling design and should not be interpreted as direct comparisons of method performance.

We assumed closure within the sampling window because it fell within known breeding dates for most focal species in Tennessee (15 April–31 May); however, we acknowledge that population closure may not be met (e.g., Fournier et al. 2022) given some focal species are migratory and detections may have included transient individuals. Consequently, occupancy estimates from single-season models should be interpreted as patterns of site use during the sampling period rather than strict occupancy, particularly with respect to absolute values. In contrast, the dynamic occupancy framework used in objective 2 allows occupancy state to vary across survey periods and is therefore more appropriate for systems where site use may change because of migration.

Objective 2: Habitat associations

To evaluate local and landscape associations of secretive marsh birds in Tennessee, we fit dynamic occupancy models to our PC + ARU encounter history data. The dynamic occupancy framework allowed occupancy state to vary across primary survey periods, thereby relaxing the closure assumption required by single-season occupancy models and providing a more appropriate framework for systems where site use may change because of migration or other temporal processes (MacKenzie et al. 2006). For local and landscape variables, we collected data on vegetation within a 25-m radius of each survey point when we deployed ARUs (Table 1). We chose the 25-m radius instead of the standard 50-m radius because of the smaller size of Tennessee wetlands and to limit disturbance at the point. We used the Braun-Blanquet approach within 5 1-m^2 quadrats spaced 5 m apart along 4 transects, in each cardinal direction (Braun-Blanquet 1932). We placed quadrats 5 m apart instead of the standard 10 m because of the small area of marshes in Tennessee. Within each quadrat, we quantified the percent cover of primary vegetation and cover classes including non-persistent emergent, persistent emergent, open water, woody or scrub, and floating aquatic vegetation cover. For landscape-scale variables, we extracted percent cover of palustrine marsh, emergent vegetation, open water, woody or shrub, agriculture, developed, and artificial flooding within 500-m buffers around each survey point (Table S2). Data sources included the NLCD, NWI, Landfire National Vegetation Classification, and Google Earth Engine (Google, Mountain View, CA, USA; Table S2). We selected a 500-m buffer given this scale has been identified as important in marsh bird habitat selection (Glisson et al. 2017). We included survey day as a detection covariate to account for temporal variation in detectability across the sampling season. Survey day was defined as the sequential day of sampling (1–50), aligned with calendar date across years, and was included as a continuous covariate in detection models. Treating survey day as a continuous covariate allowed us to model gradual seasonal changes in detection probability across the sampling period.

We used the combined PC + ARU dataset and dynamic occupancy models to evaluate factors that affect the probability of detection (p) and occupancy (ψ) for focal species and migrating sora. This dataset provided sufficient temporal replication to support dynamic occupancy modeling, allowing occupancy state to vary across survey periods. We used the unmarked package (Fiske and Chandler 2011, Kellner et al. 2023) to examine the extent to which local- and landscape-scale characteristics influence occupancy in Tennessee.

When inference on hypothesized relationships is the primary objective of a study, multiple authors suggest limiting or avoiding covariate selection (Tredennick et al. 2021, Arif and MacNeil 2022). Thus, given our second objective was to describe general patterns in occupancy and habitat associations of marsh birds in Tennessee, we focused model selection on detection and occupancy while keeping colonization and extinction parameters constant. Prior to model formulation, we checked variables for multicollinearity ($|r| \geq 0.6$), and if variables were correlated, we ensured that they did not appear in the same model. We centered and scaled all continuous covariates using the z transformation to help model convergence and parameter estimation. We performed model

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TABLE 1 Candidate variables for estimating dynamic occupancy of focal marsh bird species, including least bittern, king rail, American coot, common gallinule, sora, and pied-billed grebe, based on point count (PC) surveys and passive acoustic monitoring in Tennessee, USA, during 15 April–31 May 2022–2023.

Model parameter	Hypotheses	Variable	Description	Predicted relationship
Detection	Temporal characteristics	Survey day	Continuous variable representing the day of call broadcast PC survey period: 1–50	−
		Bout	Categorical representing the survey round or bout: 1–3	0
		Precipitation	Continuous. Daily precipitation averages in cm.	−
	Local conditions	Wind	Continuous. Average km/hr measured prior to survey	−
		Background noise	Ordinal variable representing the average background noise during PC surveys: 0 (low) to 5 (high)	−
		Temperature	Continuous, average temperature in Celsius	−
		Emergent vegetation	Percent cover of emergent vegetation in survey area of call broadcast surveys	+
Occupancy	Local cover	Floating-leaved vegetation	Percent cover of floating-leaved (e.g., lotus, lemna) vegetation in survey area of call broadcast surveys	+
		Open water	Percent cover of open water in survey area of call broadcast surveys	0
		Water depth	Measured at survey point, water depth in cm	−
		Agriculture 500 m	Planted agricultural fields consisting of crops or hay/pasture based on the National Land Cover Database (NLCD)	−
	Landscape cover	Emergent 500 m	Areas with perennial herbaceous vegetation and soils periodically saturated by water, NLCD	+
		Open water 500 m	Areas of open water with <25% vegetation cover, NLCD	0
		Palustrine 500 m	Inland wetlands based on the National Wetlands Inventory	+

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selection separately for each species. We first evaluated candidate models for detection and used the top detection model to evaluate candidate models for occupancy.

We developed 11 *a priori* candidate models representing factors that could affect the probability of detecting occupancy of focal marsh bird species (Table 1). Then, we developed 12 *a priori* candidate models with a combination of local and landscape covariates previously identified as influential for focal marsh bird occupancy (Table 1). We fit a model including local vegetation and inundation conditions that are typically associated with marsh birds, including the cover of emergent vegetation, floating-leaved aquatic vegetation, open water, and water depth. Additionally, given the importance of landscape cover in predicting marsh bird occupancy in previous studies (Bradshaw et al. 2020, Stevens and Conway 2020), we also included a landscape model with the proportion of palustrine, emergent, open water, and agricultural cover within 500 m of the survey location. Lastly, we fit a global model with local vegetation and inundation conditions and landscape cover, we included a null model without parameters, and we ran each variable individually as a single additive effect model. Similarly to objective 1, we also evaluated site as a random effect but did not include it in the final models due to poor convergence. We implemented model selection using Akaike's Information Criterion corrected for small sample sizes (AIC_c; Burnham and Anderson 2002). We present predictions for variables whose 85% confidence intervals did not overlap 0 (Arnold 2010). We completed all analyses in program R version 4.3.1 (R Core Team 2023).

RESULTS

We ground-truthed 160 locations identified as potential sites suitable for marsh birds; however, only 78 survey points met our *a priori* site selection criteria (i.e., ≥50% emergent vegetation cover). We conducted 123 PC surveys at 41 wetland survey points in western Tennessee in 2022. The most common species detected during PC surveys in western Tennessee in 2022 was American coot (48% of surveys), followed by least bittern (27% of surveys), sora (15% of surveys), and pied-billed grebe (10% of surveys). In 2023, we conducted a similar survey effort by performing 111 PC surveys at 37 wetland survey points in central Tennessee. The most common species detected during PC surveys in central Tennessee during 2023 were sora (56% of surveys), American coot (22% of surveys), pied-billed grebe (11% of surveys), and least bittern (11% of surveys). We did not detect any king rails or common gallinules during PCs.

We obtained approximately 8,190 hours of ARU recordings. The most common species detected during ARU surveys in western Tennessee during 2022 were pied-billed grebe (32% of surveys), least bittern (25% of surveys), sora (20% of surveys), American coot (19% of surveys), and common gallinule (4% of surveys). In central Tennessee during 2023, the most common species detected by ARUs included American coot (20% of surveys), sora (21% of surveys), pied-billed grebe (21% of surveys), least bittern (27% of surveys), common gallinule (10% of surveys), and king rail (1% of surveys). These values represent the proportion of survey days on which each species was detected and provide a measure of temporal detection frequency within sites rather than site-level occupancy.

There were 14 survey points where least bitterns were detected by ARUs and not by PCs. American coots were detected at 10 points by ARUs where they were not detected by PCs. Common gallinule was detected at 10 points by ARUs but was never detected by PCs. There were 25 points where pied-billed grebe was detected by ARUs and not by PCs. There were 29 points where sora were detected by ARUs and not by PCs. King rails were detected at 1 point by an ARU and never detected by PCs. We were unable to include king rails in our occupancy analyses because of the few detections.

Objective 1: Comparison between ARUs and PCs

Differences among survey approaches should be interpreted in the context of substantial differences in sampling effort and spatial detection extent. Across focal species, detection probability estimates were greater and often

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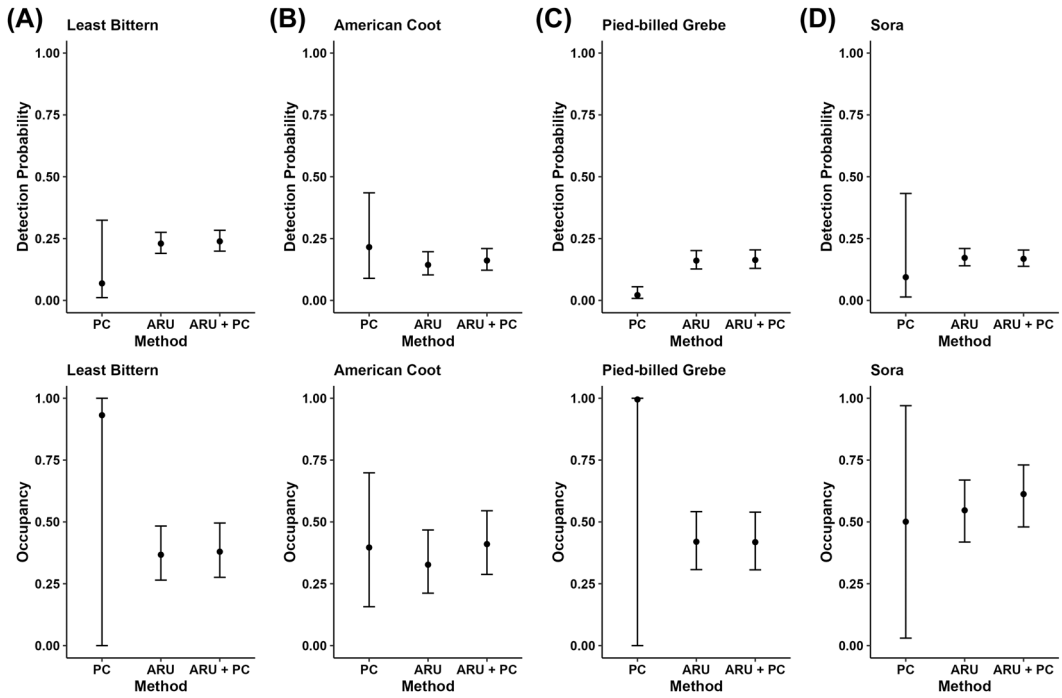


FIGURE 2 Detection probability, occupancy probability, and coefficient of variation for focal marsh bird species using 3 survey datasets fit to single-season occupancy models: point count (PC), autonomous recording units (ARU), and the integration of the 2 (ARU + PC) for focal marsh bird species in Tennessee, USA, during 15 April–31 May 2022–2023. Species are arranged by column: (A) least bittern, (B) American coot, (C) pied-billed grebe, and (D) sora. The bars represent 85% confidence intervals.

more precise for ARU-based datasets compared to PC data. The integration of ARU + PC data did not increase precision for most species compared to the ARU data alone (Figure 2).

Patterns in occupancy estimates varied among species and survey approaches, but uncertainty was consistently greatest for PC-only models. For least bittern, occupancy estimates from the PC dataset were highly uncertain, with confidence intervals spanning nearly the full parameter space (width = 1.00, CV = 85%). In contrast, ARU and ARU + PC datasets produced more precise and consistent estimates (width = 0.22, CV = 15%), with overlapping confidence intervals (Table 2). For American coot, occupancy estimates were variable across survey approaches. The PC dataset exhibited moderate uncertainty (width = 0.54, CV = 39%), while ARU and ARU + PC datasets produced more precise estimates (width = 0.26; CV = 20% and 16%, respectively). For pied-billed grebe, occupancy estimates from the PC dataset were highly uncertain (width = 1.00), reflecting poor parameter identifiability, whereas ARU and ARU + PC datasets yielded more stable estimates with narrower confidence intervals (width = 0.23) and lower CV (14–15%). Occupancy estimates for sora were variable across all datasets, but uncertainty was greatest for the PC model (width = 0.94, CV = 88%). The ARU and ARU + PC datasets produced more precise estimates (width = 0.25, CV = 11–12%). Because we had zero detections of common gallinule on PC surveys, we were unable to estimate occupancy or detection parameters for this species for objective 1.

Objective 2: Habitat associations

After checking variables for multicollinearity ($|r| \geq 0.6$), we found that for the landscape variables, the woody (500-m scales) variables were correlated ($|r| = 0.77$) with palustrine (500-m scale). At the local scale, water depth (cm) was

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TABLE 2 Beta coefficients (β) with 85% confidence intervals on logit scale, and measures of uncertainty for occupancy probability from single-season occupancy models using point count (PC), autonomous recording unit (ARU), and combined PC+ARU data sources for marsh birds in Tennessee, USA, during 15 April–31 May 2022–2023. Coefficient of variation (CV, %) and confidence interval width (width) are calculated for occupancy estimates on the probability scale using back-transformed estimates. Parameters include occupancy (ψ) and detection (*p*) probabilities.

Species	Method	Parameter	β	Lower 85% CI	Upper 85% CI	CV (%)	Width
Least bittern (*Ixobrychus exilis*)							
	PC	ψ	2.61	−15.337	20.557	85	1.00
		p	−2.61	−3.979	−1.231		
	ARU	ψ	−0.54	−0.895	−0.193	15	0.22
		p	−1.21	−1.385	−1.032		
	ARU+PC	ψ	−0.49	−0.839	−0.143	15	0.22
		p	−1.16	−1.329	−0.989		
American coot (*Fulica americana*)							
	PC	ψ	−0.42	−1.345	0.506	39	0.54
		p	−1.29	−2.049	−0.536		
	ARU	ψ	−0.72	−1.156	−0.288	20	0.26
		p	−1.78	−2.063	−1.507		
	ARU + PC	ψ	−0.36	−0.762	0.036	16	0.26
		p	−1.65	−1.886	−1.412		
Pied−billed grebe (*Podilymbus podiceps*)							
	PC	ψ	5.31	−55.905	66.523	20	1.00
		p	−3.82	−4.538	−3.099		
	ARU	ψ	−0.32	−0.683	0.037	15	0.23
		p	−1.65	−1.854	−1.450		
	ARU + PC	ψ	−0.33	−0.688	0.029	14	0.23
		p	−1.63	−1.833	−1.432		
Sora (*Porzana carolina*)							
	PC	ψ	0.004	−2.546	2.553	88	0.94
		p	−2.27	−3.733	−0.802		
	ARU	ψ	0.19	−0.191	0.568	12	0.25
		p	−1.57	−1.751	−1.391		
	ARU+PC	ψ	0.46	0.062	0.853	11	0.25
		p	−1.60	−1.772	−1.426		

correlated with woody vegetation ($|r| = 0.62$), and these variables were not included in the same model. After following our hierarchical approach to model selection, the final model included survey day for probability of detecting occupancy for all focal marsh bird species (Table 3 and S3). However, for least bitterns, common gallinule, and sora, the relationship was uncertain (i.e., 85% confidence intervals contained 0; Table 4). For each

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TABLE 3 Top dynamic occupancy model for each focal marsh bird species for estimating occupancy (ψ), colonization (γ), extinction (ε), and detection (*p*) of marsh birds in Tennessee, USA, during 15 April–31 May 2022–2023. To select the top model, we first selected the top detection model and then conducted model selection for occupancy using 12 *a priori* models. We did not conduct model selection for colonization or extinction and elected to set these model components as constant.

Species	Model
Least bittern (*Ixobrychus exilis*)	ψ(.)γ(.)ε(.)*p*(survey day)
Common gallinule (*Gallinula galeata*)	ψ(floating aquatic vegetation + agriculture 500 m + emergent 500 m)γ(.)ε (.) *p*(survey day)
American coot (*Fulica americana*)	ψ(.)γ(.)ε (.)*p*(survey day)
Pied-billed grebe (*Podilymbus podiceps*)	ψ(.)γ(.)ε(.)*p*(survey day)
Sora (*Porzana carolina*)	ψ(open water 500 m)γ(.)ε(.)*p*(survey day)

additional week throughout the season, we were 0.7 (85% CI = 0.50–0.93) times less likely to detect American coot (Figure 3). For pied-billed grebe, we were 0.8 (85% CI = 0.66–0.87) times less likely to detect one for each additional week (Figure 3).

Occupancy estimates across sites varied across focal species. For least bittern, American coot, and pied-billed grebe, the null model was the best for occupancy (Table 3), and initial site occupancy was 18% (85% CI = 0.12–0.26), 26% (85% CI = 0.18–0.36), and 28% (85% CI = 0.21–0.37), respectively (Table 4). For common gallinule, 3 models were competitive (ΔAIC < 2), and conditional model averaging indicated that site occupancy increased with floating aquatic vegetation and agricultural cover (500 m). For every 10% increase in floating aquatic vegetation cover at a site, common gallinule were 1.08 times more likely to occupy the site (85% CI = 1.02–1.15; Table 4). For every 10% increase in agricultural cover (500 m), common gallinule were 1.11 times more likely to occupy a site (85% CI = 1.03–1.20). The final model for sora included open water at the landscape scale (500 m). For every 10% increase in open water at a site, sora were 1.15 times more likely to occupy a site (85% CI = 1.06–1.25; Table 4).

DISCUSSION

Monitoring of cryptic species requires robust survey designs that provide reliable population estimates, while accounting for imperfect detection, and lead to informed conservation and management actions. In our study, differences in probability of detecting occupancy and occupancy estimates among survey approaches were substantial; however, these differences are best understood as consequences of sampling design rather than inherent differences in method performance. The ARUs provided substantially greater temporal sampling effort than PCs, which resulted in higher detection probabilities and, in some cases, more precise occupancy estimates. Integrating ARU and PC data did not substantially improve estimates of detection, except for the American coot, which is commonly detected more visually than auditorily. For occupancy, integrating ARU and PC data improved precision. Our results are consistent with previous research showing that traditional survey methods such as PCs can yield low detection probabilities for secretive marsh birds, particularly under limited temporal sampling, which may constrain inference (Tegeler et al. 2012, Sidie-Slettedahl et al. 2015). The use of passive acoustic monitoring significantly increased temporal sampling effort, which in turn resulted in higher detection rates and, in some cases, more precise occupancy estimates, supporting findings from similar studies across other taxa (Drake et al. 2021). Collectively, our study adds to a growing body of literature demonstrating that increasing temporal sampling effort through passive acoustics can improve inference in low-detection systems, while also emphasizing the importance of aligning survey design with the ecological processes being studied.

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TABLE 4 Beta coefficients and 85% confidence intervals for covariates related to occupancy (ψ) and detection (*p*) of marsh birds in Tennessee, USA, during 15 April–31 May 2022–2023. Asterisks (*) indicate model averaging of top models (difference in Akaike's Information Criterion < 2). We did not conduct model selection for colonization (γ) or extinction (ε) and elected to leave them as constants.

Species	Parameter	Variable	β	Lower 85% CI	Upper 85% CI
Least bittern (*Ixobrychus exilis*)					
	ψ	Null	−1.51	−2.01	−1.03
	p	Survey day	0.008	−0.01	0.03
Common gallinule (*Gallinula galatea*)*					
	ψ	Floating aquatic vegetation	0.81	0.22	1.40
		Agriculture 500 m	1.02	0.27	1.78
		Emergent 500 m	−2.06	−4.66	0.54
	p	Survey day	0.017	−0.02	0.05
American coot (*Fulica americana*)					
	ψ	Null	−0.95	−1.41	−0.49
	p	Survey day	−0.05	−0.10	−0.01
Pied-billed grebe (*Podilymbus podiceps*)					
	ψ	Null	−0.93	−1.32	−0.53
	p	Survey day	−0.04	−0.06	−0.02
Sora (*Porzana carolina*)					
	ψ	Open water 500 m	1.38	0.56	2.21
	p	Survey day	−0.02	−0.04	0.001

Because several focal species are migratory, many detections likely represent transient individuals, and closure assumptions of single-season occupancy models were likely violated. Consequently, estimates from objective 1 should be interpreted as patterns of site use during the sampling period rather than true occupancy. The use of dynamic occupancy models in objective 2 partially mitigates this issue by allowing occupancy state to vary across survey periods, providing a more appropriate framework for temporally dynamic systems. Additionally, within wetland sites we randomly selected 1-3 survey points depending on wetland size, resulting in multiple subsamples within some sites. We were unable to include a site level random effect due to poor model convergence, likely because some sites had only one survey point and therefore provided limited information. Consequently, the final models did not include a site random effect, so potential non-independence of points within sites may have resulted in overconfidence in statistical differences. Results should be interpreted with this potential limitation in mind.

We provide further empirical support that ARUs can supplement marsh bird sampling approaches; however, improvements in detection must be balanced against additional logistical and financial considerations. Relative to traditional PC surveys, ARU-based monitoring requires the purchase and maintenance of recording equipment, field time for deployment and retrieval, and substantial data storage, processing, and validation efforts. Consequently, the benefits of increased temporal sampling effort and improved detection may not always outweigh the additional resources required to implement acoustic monitoring, particularly for long-term programs operating under limited budgets or personnel constraints. For instance, Drake et al. (2021) found PCs did not add much value to ARU survey data when surveying avian biodiversity, but the cost and time of using only ARU data did not outweigh the benefits

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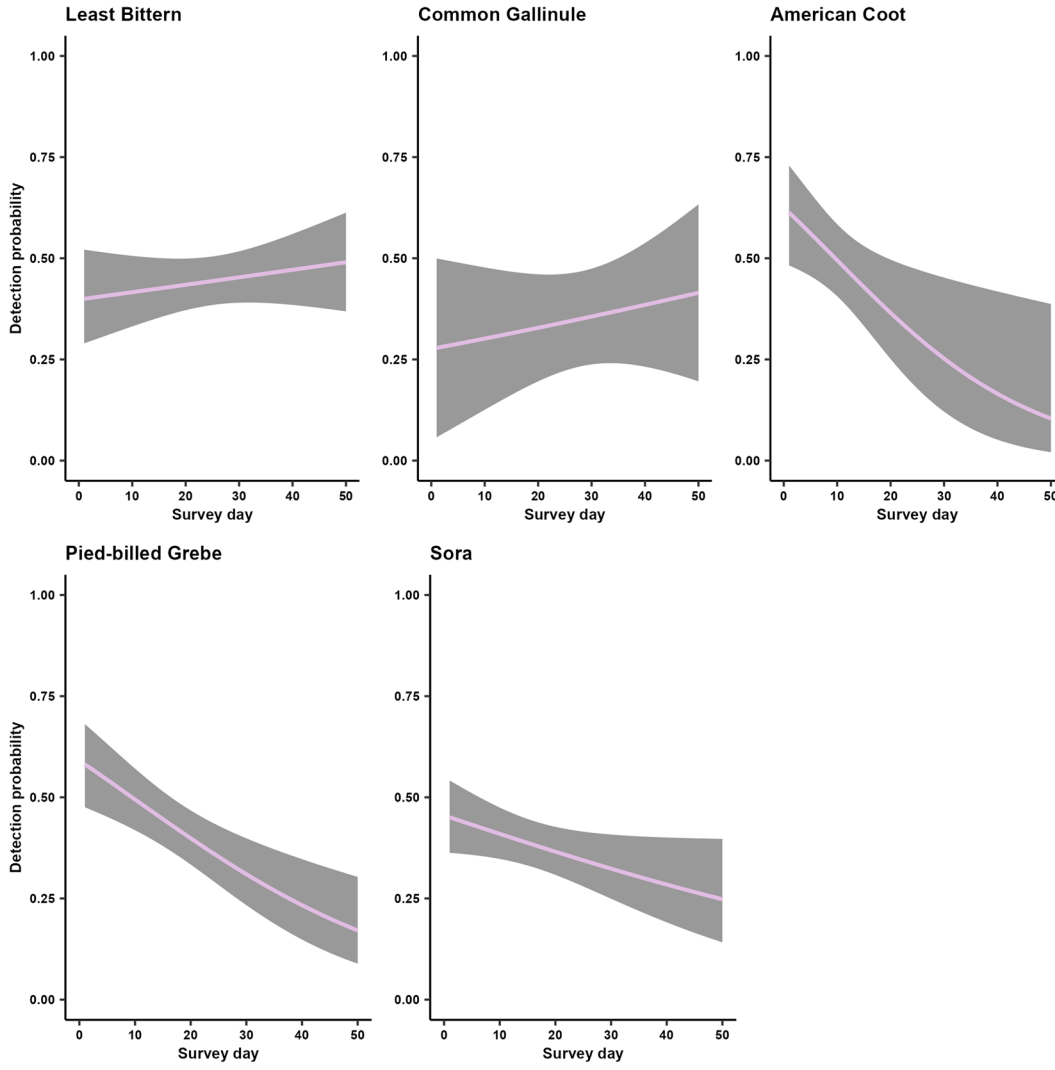


FIGURE 3 Relationship between detection probability and survey day for focal marsh bird species in Tennessee, USA, from dynamic occupancy models, during 15 April–31 May 2022–2023. Shaded areas represent 85% confidence intervals.

of the combined approach. Despite current limitations, recent advancements in bioacoustics analysis software, including improvements to platforms such as BirdNET and OpenSoundscape, suggest automated processing may improve the feasibility of broader implementation (Kahl et al. 2021, Lapp et al. 2023). However, scaling ARU-based monitoring to regional or continental programs would require addressing equipment costs, deployment logistics, data management, processing capacity, and coordination among observers. As such, ARUs may be most immediately applicable in targeted monitoring efforts or as a complement to existing survey programs rather than a direct replacement at large spatial scales.

Across focal species, landscape-scale factors most influenced marsh bird site occupancy, whereas local-scale vegetation was rarely influential. However, our observation that landscape-scale factors outweighed local vegetation characteristics contrasts with several foundational studies. For instance, Glisson et al. (2015) found that local vegetation structure strongly predicted marsh bird occurrence in Wisconsin wetlands, while our Tennessee sites showed minimal influence of local vegetation metrics. This difference might reflect the highly

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altered state of Tennessee's wetlands, where broader landscape context may limit the influence of site-level characteristics. Additionally, local habitat measurements were collected at the survey point and used to model initial occupancy in a dynamic framework and thus may not fully represent spatial or temporal variability in conditions across the broader survey area, particularly for dynamic variables such as water depth in managed wetlands. Likewise, the detection probabilities we observed for most species between survey periods suggest wetland use may be more dynamic than previously recognized. This result supports recent work by Fournier et al. (2022) where coauthors questioned the feasibility of closure assumptions in marsh bird occupancy models. Accordingly, habitat associations presented here should be interpreted as relationships with site use rather than strictly breeding occupancy, particularly in systems where occupancy state may change across the sampling period. Collectively, marsh bird habitat selection may be more context-dependent than earlier research suggested, highlighting the need to consider regional differences when applying habitat management guidelines.

Tennessee wetlands are highly modified, and only an estimated 31% of the state's wetlands remained by the 1980s (Dahl 1990, Johnson 2007). Many remaining wetland areas are impounded and managed explicitly for waterfowl. As such, starting in early March, wetland managers typically removed water and dense persistent emergent vegetation from impoundments to prepare for planting crops or to conduct moist-soil management (Winstead and King 2006). A previous study by Bradshaw et al. (2020) found marsh bird occupancy increased at intermediate levels of waterfowl management intensity including in wetlands with moist-soil management and food plot management, both common approaches in Tennessee. However, we suspect the intensity of management for waterfowl was likely greater in Tennessee than in the study area of Bradshaw et al. (2020) in Illinois (A. G. Blake-Bradshaw, University of Illinois Urbana-Champaign, personal observation). Highly managed wetlands, such as those where persistent emergent vegetation is removed, likely do not support both breeding marsh birds and migratory or wintering waterfowl equally. Most of our sites are primarily managed for waterfowl, suggesting that current management practices, in Tennessee at least, may not provide sufficient quality and quantity of habitat for most secretive marsh birds during spring migration and breeding.

Landscape-scale habitat dynamics influence marsh bird distributions, yet most research has focused on local-scale characteristics (Stevens and Conway 2020). Our findings suggest that landscape-scale factors may be more influential than local habitat metrics measured in this study, particularly in Tennessee, where wetland connectivity and availability may be insufficient to support breeding marsh bird populations. However, this pattern may also reflect limitations in measuring local habitat characteristics in marsh systems, where conditions are difficult to quantify consistently across the full survey area and may not be well represented by point-based measurements. The temporal patterns in our data suggest most species primarily use Tennessee's wetlands as migratory stopover points rather than breeding grounds. For example, sora, and most other species, showed high detection rates in early April that declined sharply over time, indicating transient use during migration. Only least bitterns maintained consistent numbers across survey periods, though we confirmed just one breeding pair. This pattern of wetland use during migratory periods aligns with observations from Malone et al. (2023) that limited wetland availability in mid-latitude states may constrain both migratory and breeding marsh birds. The consistent decline in detections after the first survey round (15–30 April) suggests Tennessee's wetlands may lack the scale or connectivity needed to retain breeding populations. Understanding these patterns will require future research into habitat selection and migration chronology of secretive marsh birds to better characterize how marsh birds use Tennessee's wetland network during different phases of their annual cycle.

MANAGEMENT IMPLICATIONS

Long-term, continuous monitoring using passive acoustic monitoring may provide a more efficient way for managers to collect data on secretive marsh birds, understand their distributions, and assess the efficacy of management actions for these increasingly rare species. Given that landscape-scale factors were stronger predictors of

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marsh bird site use than local vegetation characteristics in our study, and that most species appeared to use Tennessee wetlands primarily during migration, management actions may benefit from emphasizing broader habitat availability and connectivity. Managers wanting to promote marsh bird use and support marsh bird populations in mid-latitude states may consider maintaining some wetlands with prolonged inundation and encourage perennial emergent vegetation growth through the spring and summer to provide sufficient cover and food during migration and potential breeding periods. However, our results suggest that intensive management of small, isolated wetlands may provide limited benefits relative to strategies that enhance wetland complexity and connectivity at larger spatial scales.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ETHICS STATEMENT

All data collection was conducted under required Tennessee Wildlife Resources Agency and U.S Fish and Wildlife Service permits.

DATA AVAILABILITY STATEMENT

Data are available at https://zenodo.org/records/19924154.

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